

Toward Task-Aware Reuse of Cellular Molecular Receivers: History-Aware Prediction on Experimental FGF2–ERK Data

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Abstract—We ask which observable information predicts a cellular molecular receiver’s response to repeated chemical commands and whether it can shorten the wait between commands. On public experimental FGF2–ERK single-cell trajectories, a nested post hoc comparison with a fixed, previously calibrated mechanistic reference (the authors’ B3 model) showed that, on a protocol unseen in training, reporter filtering reduced the 10-min-ahead error of a current-state predictor by 9.1%, the B3 increment reduced the error of the filtered predictor by 4.5% and that of a lagged non-mechanistic predictor by 6.9%, and explicit input-history features added only 1.0%; in the evaluated next-command windows the mechanism raised the aggregate error. In simulation on B3, feedback-driven waiting rules completed a fixed probe task 4.8 min faster than fixed waits that already use the known previous command; with every rule calibrated to a shared target of 0.92, the gap was 6.1 min at equal success and held in all 20 calibration/evaluation splits. A smoothed reporter matched the history-aware estimator at the reference noise, the latter helped at higher noise, and under correlated noise its reliability recovered only with a likelihood that models the correlation, and not in every split. No result concerns measured task reliability.

Index Terms—Molecular communication, cellular receivers, mechanism-informed prediction, receiver reuse, waiting rules, FGF2, ERK.

I. INTRODUCTION

Molecular communication (MC) with living receivers asks a chemical command to produce a specified response in a cell. Reception then means executing the requested response on time, and a previous command can change the next response through delivery persistence, receptor dynamics, intracellular feedback, or cellular state. For repeated tasks the operational question is when the receiver can be used again at matched reliability. We ground this question in recorded biology: FGF2 stimulation of PC-12 cells with ERK activity read by a FRET reporter [1], treated as a continuous receiver output without retroactive symbols or cell-fate labels.

Two lines of work bound the claim. In cell dynamics, the FGF2 data were used to infer signaling architecture, including feedback and a nontrivial washout response [1]; stimulation frequency shapes ERK responses [2]; automated chemical feedback has controlled mammalian signaling [3]; and a learned dynamical model enabled model predictive control of thousands of single cells [4]. In MC, memory is classical: diffusion and ligand binding produce intersymbol interference and reception noise [5]–[7], which guard intervals, ISI-aware

modulation and adaptive threshold detection mitigate [8], [9]. Learning-based and semantic MC [10]–[12], experimental and engineered biological receivers [13]–[15], error control under cellular noise [16] and long-term experimental MC with its reuse tradeoffs [17] extend this view. The open question is which information makes a *next task* predictable for a living receiver, whose memory lies in its biochemistry rather than only in the channel, and how much a waiting decision gains from it.

This paper makes two bounded contributions. (i) On experimental FGF2–ERK trajectories, a nested comparison quantifies in which forecast phases reporter filtering, a fixed calibrated mechanistic increment and explicit input-history features add predictive value, against a lagged non-mechanistic baseline and with leave-one-protocol-out (LOPO) validation in which predictor tuning is restricted to the training protocols. (ii) In simulation on B3, for a fixed probe task, we quantify how much feedback and history estimation add to fixed waits that already use the known previous command, under shared calibration targets, and how the gain depends on the observation-noise level and model, with no-probe, bandwidth and correlated-noise checks. The two parts are separate evidence streams (Fig. 1): the experimental predictors do not enter the simulated waiting rule.

II. SYSTEM AND TASK MODEL

Let $u(t) \geq 0$ denote the commanded FGF2 concentration in ng/ml. Delivery to a receiver is represented abstractly as

$$\begin{aligned} c_R(t) &= \mathcal{T}_\theta[u](t), & \dot{x}_j &= f_{\theta_j}(x_j, c_R), \\ z_j &= g(x_j), & y_{j,i} &= z_j(t_i) + \nu_{j,i}, \end{aligned} \quad (1)$$

where c_R is the local concentration, x_j the unobserved state of cell j , z_j its noise-free normalized reporter, and $y_{j,i}$ the observation with noise $\nu_{j,i}$ at frame t_i . The operator $\mathcal{T}_\theta$ can include delivery lag and washout; neither it nor c_R is calibrated here, and the FRET ratio is not a molecular count or a cell-fate measurement. The available information $\mathcal{H}_t$ contains the executed commands and the observations whose feedback has arrived by t ; offline, each observation is assumed available at its recorded time.

We fix the amplitude and duration of the next command and decide only when to send it. After a previous command ending at time 0, a 5-min probe at the same amplitude is

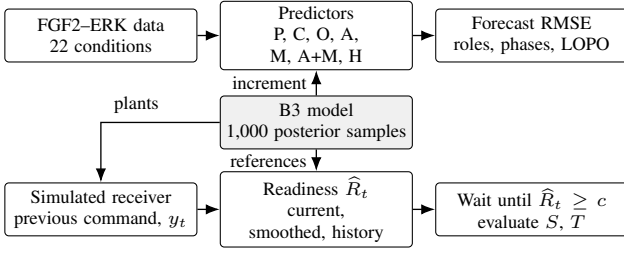


Fig. 1. Two evidence streams. Top: experimental trajectories train and test the predictors, with B3 as a fixed, previously calibrated mechanistic increment. Bottom: B3 posterior samples act as simulated receivers and as estimator references for the waiting rule (Sec. V-C). The experimental predictors do not enter the waiting rule; delivery and latent states are not calibrated.

sent at $\tau \in \mathcal{T} = \{0, 2, \dots, 120\}$ min. With z_j^τ the noise-free response when the probe is sent at τ after the recorded history, and $r_{\text{naive},j}$ the same receiver's rise to the probe without a previous command, the task outcome and readiness are

$$S_j(\tau) = \mathbf{1} \left\{ \max_{s \in [\tau, \tau+W]} z_j^\tau(s) - z_j^\tau(\tau) \geq \kappa r_{\text{naive},j} \right\}, \quad (2)$$

$$R_{j,t} = \mathbb{P}(S_j(t) = 1 \mid \mathcal{H}_t),$$

with task window $W = 20$ min and $\kappa = 0.5$, a prescribed probe criterion rather than one derived from a biological task (W and the penalty D below are likewise prescribed assessment settings). Observation noise affects the feedback used for decision-making; success is scored from the noise-free model response, and decisions use only observations available by τ . The counterfactual $r_{\text{naive},j}$ is a simulation benchmark; an experiment would have to measure or estimate it, or replace it by a fixed measurable threshold. A causal check replaces the rise by the probe-induced increment $\Delta z_j(s; \tau) = z_j^\tau(s) - z_j^\emptyset(s)$, with $z_j^\emptyset$ the response without the probe, and counts success when $\max_{s \in [\tau, \tau+W]} \Delta z_j(s; \tau) \geq \kappa r_{\text{naive},j}$. The failure-penalized completion time is $T = \tau + W$ (the end of the assessment window, not the time the criterion is first met) on success and $D = 140$ min on failure; the design objective is to minimize $\mathbb{E}[T]$ subject to $\mathbb{P}(S = 1) \geq q = 0.9$, an empirical target rather than a coverage guarantee. The experimental data have no task labels, so Eq. (2) is evaluated only in simulation.

III. MECHANISM-INFORMED PREDICTION AND WAITING

A. Nested Predictors

Separate predictors are fit for horizons $h = 2$ and 10 min after a fixed 10-min observed warmup. Every learned predictor receives the same known future command (its exposure in 2-min bins, the concentration at $t + h$, and h); future reporters are targets only. The current-state set C contains $u_t, u_t^2, y_t, y_t^2, u_t y_t$, the 2-min slope $(y_t - y_{t-2})/2$ (so C is not memoryless) and the future-command features. Causal filters with scales $\tau \in \{2, 10, 60\}$ min summarize input and reporter history,

$$q_\tau(t) = \frac{1}{\tau} \int_0^t e^{-(t-s)/\tau} u(s) ds, \quad (3)$$

$$r_{\tau,i} = e^{-\Delta_i/\tau} r_{\tau,i-1} + (1 - e^{-\Delta_i/\tau}) y_i,$$

with $r_{\tau,0} = y_0$ and $\Delta_i = t_i - t_{i-1}$; the scales are computational choices, not identified rate constants. O adds the reporter filters r_τ to C. As a non-mechanistic history baseline, A adds to C the reporter and command values of the previous $L \in \{5, 15, 30\}$ frames (10–60 min); before the first frame the reporter lag repeats that frame and the command lag is the known command (zero before onset), which affects origins before 30 min when $L = 15$ is selected (2-min A, LOPO) but not the $L = 5$ of the main 10-min analysis. As a mechanism we execute the authors' B3 model with their 1,000 posterior parameter samples. With $f_s(t)$ the noise-free reporter of sample s under the recorded command, the samples are weighted by each cell's past reporter,

$$w_s(t) \propto \exp \left[- \sum_{t_i \leq t} (y_i - f_s(t_i))^2 / (2\sigma^2) \right], \quad (4)$$

$$\hat{\delta}_{t,h} = \sum_s w_s(t) [f_s(t+h) - f_s(t)],$$

a fixed-parameter particle approximation of the B3 belief that uses the same information as the learned predictors. M adds $\hat{\delta}_{t,h}$ to O, A+M adds it to A, and H adds the input filters q_τ and the cumulative exposure $\int_0^t u(s) ds$ to M. The B3 increment already uses the command history and past observations, so H tests only the additional value of explicit input-history features. Persistence P ($\hat{y}_{t+h} = y_t$) needs no training. Each set is fit by ridge regression with equal weight per condition and standardization from training rows; the penalty (from 10^{-8} to 10^4), σ (seven values) and L are selected on validation.

B. Calibrated Waiting Rules

A waiting rule probes at

$$\tau = \begin{cases} \inf \{t \in \mathcal{T} : \hat{R}_t \geq c\}, & \text{if such a } t \text{ exists,} \\ \max \mathcal{T}, & \text{otherwise.} \end{cases} \quad (5)$$

Three estimators of R_t are compared. The current estimator uses only y_t , through a Gaussian kernel over the (reporter, success) states of reference receivers; the smoothed estimator applies the same kernel to a causal exponential smoothing of the observations (scale 2–16 min); the history-aware estimator weights reference receivers under the known command history by all observations so far, as in Eq. (4), with likelihood s.d. equal to the noise s.d. floored at 0.001 (a floor of 0.005 is a pre-specified variant; the floor regularizes the estimator and is not an assumed noise level). Reference receivers are the calibration samples (leaving out the receiver itself during calibration); evaluation samples never enter references, thresholds or waits. Because per-frame calibration does not guarantee the reliability of an adaptive stopping rule, the whole rule is calibrated: for the current and history estimators we take the first threshold on the grid $\{0.5, \dots, 0.99\}$ whose calibration success reaches q , a heuristic guided by the completion-time objective; for each smoothing scale we take a feasible threshold the same way and then the scale with the smallest calibration mean completion; if nothing is feasible, the largest threshold is used and the rule is reported as missing the target. As the adaptive rules know the previous command, the main fixed comparator also uses it: two waits (τ_3, τ_{30}) are chosen jointly over the

TABLE I
PROTOCOLS, ROLES AND SINGLE-CELL TRAJECTORIES (22 CONDITIONS,
1,565 CELLS, 2-MIN FRAMES). B3 USE: CONCENTRATIONS WHOSE MEANS
THE AUTHORS FITTED (FIT) OR USED IN THEIR ARCHITECTURE
COMPARISON (ARCH.).

Protocol	Role	ng/ml	Cells	B3 use
Sustained	Train	0.25, 2.5, 25, 250	248	fit 2.5, 250
3/20 pulses	Train	2.5, 25, 250	207	fit 2.5, 250
Single 5 min	Val.	0.25, 2.5, 25, 250	370	arch. 2.5, 250
Single 10 min	Test A	0.25, 2.5, 25, 250	248	—
Single 60 min	Test B	2.5, 25, 250	167	—
Mixed	Test C	0.25, 25	171	—
Mixed	Viewed	2.5, 250	154	arch. 2.5, 250

61^2 grid pairs to minimize the equal-weight calibration mean completion subject to pooled calibration success $\geq q$; a variant requires q for each history, and a single wait for both histories is also reported. Choosing the fastest feasible threshold instead of the first and a finer threshold grid are sensitivity analyses, and a shared calibration target of 0.92 or 0.94 for every rule, with evaluation still against 0.9, tests whether a comparator at the edge of the target drives the comparison. Paired 95% bootstrap intervals resample evaluation posterior samples (2000 resamples), keeping both histories and all rules paired.

IV. DATA AND EVALUATION PROTOCOL

A. Data and Roles

All numerical files come from the paper-linked repository at a pinned commit [18], [19], and commands follow the authors' model XML. Model time 0 is the stimulation onset, and per-cell normalization uses only earlier frames (a mean over raw 2–38 min in the exports and a median over raw 10–30 min, reconstructed by us, for the mixed protocol whose origin is raw 42 min), so every forecast uses information available at its origin. Table I lists the 22 conditions. The roles were frozen before any newly used response file was downloaded: Test A is a new protocol, Test B's $[0, 60]$ timing was inferred from the authors' B3 simulations (± 1 min sensitivity), and Test C uses new concentrations recorded in the session of the viewed mixed blocks, on which an earlier pilot was tested. Every analysis below was frozen in its own protocol before running, but all are post hoc with respect to inspected data.

B. Mechanism, Validation and Scores

B3 is executed by parsing the authors' model files; no equation is retyped, no parameter is refit, and our posterior-predictive mean matches their exported predictions within 8.5×10^{-6} . Because the authors fitted B3 to the sustained and 3/20 means and compared architectures on the single 5-min and viewed mixed blocks at 2.5 and 250 ng/ml (Table I), M, A+M and H use a calibrated reference, not a model blind to those protocols. In LOPO each of six protocol groups is held out in turn, with predictor tuning restricted to the training protocols (σ on the training groups; penalty and L by inner leave-one-group-out) while B3 remains fixed; only the 10-

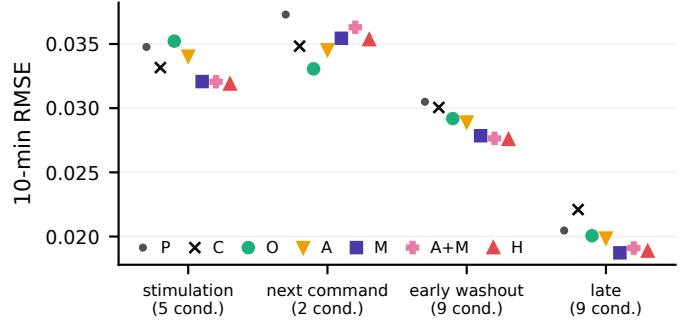


Fig. 2. Condition-equal 10-min RMSE on Tests A–C by forecast phase for all predictors (axis not starting at zero).

and 60-min pulse protocols were absent from the original B3 fitting and architecture comparison. Forecast rows take the first matching phase in the order stimulation (command on at t), next command (a pulse starts in $(t, t + h]$), early washout (last pulse ended at most 30 min before t), and late; after its warmup Test A contains only washout rows. The score is the square root of the equally weighted condition MSE,

$$\text{RMSE}_{\text{cond}}(h) = \left[\frac{1}{|D|} \sum_{d \in D} \frac{1}{N_{d,h}} \sum_{i \in d} (\hat{y}_{i+h} - y_{i+h})^2 \right]^{1/2}, \quad (6)$$

reported with per-condition directions. Rows and cells are not independent interventions and experimental-day identifiers are unavailable, so no significance test is reported.

V. RESULTS AND DISCUSSION

A. Nested Prediction

Table II(a) reports the nested predictors. On Test A at 10 min, reporter filtering reduced the error by 9.1% relative to C, the B3 increment by 4.5% relative to O, and explicit input-history features by 1.0% relative to M, giving 0.02098, 8.2% below persistence; these reductions use different baselines and are not additive. The H-versus-M gain was small and not consistent across conditions (two of four on Test A; 0.4% and 0.1% on B and C). The lagged baseline A was 2.3% worse than O on Test A (better on B and C), but adding the B3 increment to it still reduced the error by 6.9% on Test A (four of four concentrations), 3.6% on B and 4.2% on C, and H and A+M were within 1.4%. The mechanistic increment thus adds value beyond a directly represented input and reporter history, whereas explicit input history adds little once the mechanism is present.

B. Forecast Phases and LOPO

By phase (Fig. 2), the B3 increment reduced the error of O in every phase except the next-command window (by 8.9% during stimulation, 4.6% in early washout and 6.6% late), and that of A in the same phases. During stimulation the reporter filters hurt (O 0.03522 versus C 0.03316). The next-command window is covered by two mixed conditions with ten forecast origins each: the increment helped at 0.25 ng/ml (O 0.03652, M 0.03485; A 0.03774, A+M 0.03601) but raised

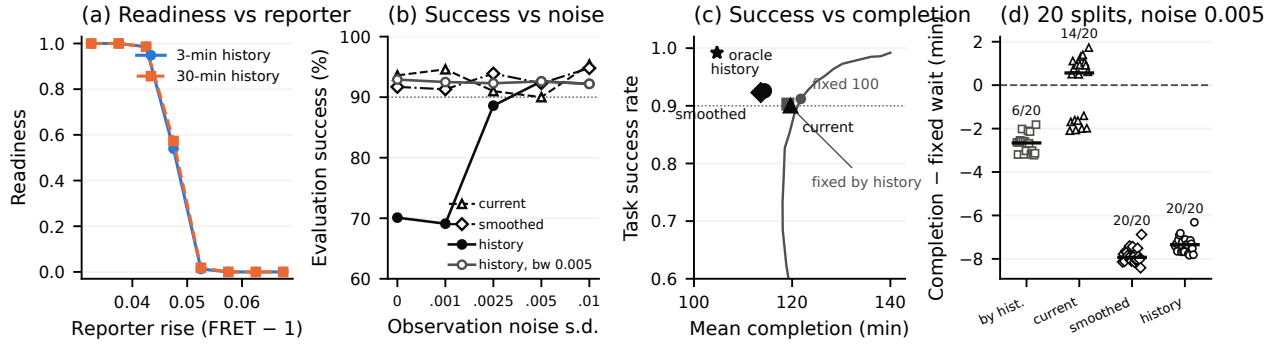


Fig. 3. Calibrated waiting on the authors' B3 model (25 ng/ml; task: probe rise $\geq 50\%$ of the naive rise within 20 min). (a) Readiness versus the current noise-free reporter after 3- or 30-min commands. (b) Evaluation success versus observation noise for the current, smoothed and history rules (likelihood s.d. floored at 0.001) and the history rule with floor 0.005; dotted: target 0.9. (c) Success versus mean failure-penalized completion at noise 0.005 on the primary split; gray curve: single fixed waits; square: history-conditioned fixed waits. (d) Twenty splits with fresh noise: completion minus that of the single fixed wait (bars: medians; numbers: splits reaching 0.9 evaluation success). Thresholds and waits chosen on calibration samples; one simulated model structure, not measured cells.

TABLE II

CONDITION-EQUAL 10-MIN RMSE UNLESS NOTED. (A) VAL.: SINGLE 5-MIN PULSE; A: SINGLE 10-MIN PULSE; B: SINGLE 60-MIN PULSE (INFERRED TIMING); C: MIXED PULSES AT 0.25 AND 25 NG/ML. P AND C ARE PRE-REGISTERED; O, A, M, A+M AND H ARE POST HOC. BOTTOM ROWS COUNT CONDITIONS WITH LOWER MSE. (B) CORRECTED LOPO WITH PREDICTOR TUNING RESTRICTED TO THE TRAINING PROTOCOLS; B3 USE: THE 2.5 AND 250 NG/ML CONDITIONS WERE FITTED (FIT) OR USED IN THE ARCHITECTURE COMPARISON (ARCH.; °: OTHER CONCENTRATIONS). BOLD: SMALLEST PER COLUMN (A) OR ROW (B).

(a) Fixed roles					
Model	Val.	A	B	C	A, 2 min
P: persistence	0.02353	0.02287	0.02651	0.02876	0.01302
C: current + slope	0.02503	0.02442	0.02715	0.02825	0.01342
O: C + reporter filters	0.02403	0.02220	0.02715	0.02827	0.01283
A: C + lags	0.02343	0.02272	0.02565	0.02769	0.01300
M: O + B3 increment	0.02197	0.02120	0.02448	0.02658	0.01263
A+M: A + B3 increment	0.02161	0.02114	0.02474	0.02652	0.01270
H: M + input history	0.02195	0.02098	0.02439	0.02656	0.01260
H beats M	1/4	2/4	2/3	1/2	2/4
M beats O	4/4	3/4	2/3	2/2	3/4
A+M beats A	4/4	4/4	3/3	1/2	4/4
O beats C	4/4	4/4	2/3	1/2	4/4

(b) Leave-one-protocol-out

Held-out	O	M	H	A	A+M	B3 use
Sustained	0.02102	0.01969	0.01941	0.02025	0.01931	fit
3/20 pulses	0.03680	0.02886	0.02830	0.03287	0.02837	fit
Single 5 min	0.02243	0.02153	0.02139	0.02196	0.02134	arch.
Single 10 min	0.02084	0.02106	0.02104	0.02116	0.02082	none
Single 60 min	0.02701	0.02410	0.02369	0.02486	0.02422	none
Mixed, 0.25/25	0.02709	0.02623	0.02610	0.02706	0.02628	arch.°

the error at 25 ng/ml (O 0.02920, M 0.03603; A 0.03102, A+M 0.03659), so the pooled error rose from 0.03306 to 0.03544. The mechanistic increment thus increased the aggregate error in the evaluated next-command windows despite its benefits during washout. In the corrected LOPO (Table II(b)), H beat M in all six folds (by 0.1–2.0%) and A+M beat A in all six (by 1.6–13.7%), including both protocols unseen by B3; with the 10-min pulse held out, however, M was 1.1% worse

TABLE III

CALIBRATED WAITING ON THE PRIMARY SPLIT BY PREVIOUS COMMAND (NOISE 0.005): EVALUATION SUCCESS S AND MEAN FAILURE-PENALIZED COMPLETION T AFTER THE 3- AND 30-MIN COMMANDS. "FIXED 92/102": HISTORY-CONDITIONED FIXED WAITS (POOLED CONSTRAINT); "CAL. 0.92": THE SAME WITH CALIBRATION TARGET 0.92, AT WHICH THE HISTORY AND SMOOTHED RULES KEEP THEIR THRESHOLDS. POOLED VALUES IN THE TEXT AVERAGE THE TWO HISTORIES.

Rule	S_3 (%)	T_3 (min)	S_{30} (%)	T_{30} (min)
Current obs.	88.8	115.3	91.2	124.0
Smoothed (16 min)	92.6	108.6	92.0	118.6
History-aware	91.0	109.1	94.2	119.6
Fixed 100 min	97.4	120.5	85.0	123.0
Fixed 92/102 min	92.4	114.1	88.2	124.1
Fixed 90/106 min (each ≥ 0.9)	90.2	112.9	91.6	127.2
Fixed 94/104 min (cal. 0.92)	95.0	115.3	90.2	125.6
Oracle	100.0	99.0	98.4	110.4

than O and A+M only 0.1% better (0.02082 versus 0.02084). The sustained and 3/20 folds contain conditions used to fit B3, so their mechanistic gains are not out-of-sample for the mechanism.

C. Calibrated Waiting on B3

Readiness and waiting were simulated on B3 with 25 ng/ml commands, a 3- or 30-min previous command and observations every 2 min with noise s.d. 0.005. Readiness depended almost only on the current reporter (Fig. 3(a)); waits of 6–48 min never succeeded. Of the 122,000 (history, sample, wait) states, none met the criterion without a probe at $\kappa = 0.5$ or 0.6 (the largest no-probe rise was 47% of the naive rise), but 79 (0.065%, all after the 3-min command) did at $\kappa = 0.4$. The causal criterion agreed with Eq. (2) for 97.0–97.2% of the states at each κ and disagreed at 1.5–5.7% of the rules' stop states ($\kappa = 0.4, 0.5$), almost always by adding successes; rescoring raised every rule's success, the order by completion changed only for two pairs that ended within 0.25 min of each other, and the history rule stayed 3.4–5.4 min faster than history-conditioned fixed waits at $\kappa = 0.4$ and 0.5 under both labels.

TABLE IV

SENSITIVITY AND MISMATCH OF CALIBRATED WAITING ON THE PRIMARY SPLIT. ENTRIES: EVALUATION SUCCESS (%) / MEAN FAILURE-PENALIZED COMPLETION (MIN); WAIT: SINGLE FIXED WAIT (MIN); BY HISTORY: HISTORY-CONDITIONED FIXED WAITS; *: CALIBRATION TARGET NOT REACHED (FALLBACK). ΔT_{H-S} : HISTORY MINUS SMOOTHED COMPLETION (MIN) WITH PAIRED 95% BOOTSTRAP INTERVAL. $\dagger$: FROZEN $\kappa = 0.5$ RULES EVALUATED UNDER AR(1) OBSERVATION NOISE ($\rho = 0.8$, S.D. 0.005) OR A REPORTER OFFSET OF $+2.5 \times 10^{-3}$. $\ddagger$: THRESHOLDS RE-CHOSEN ON AR(1) CALIBRATION SAMPLES WITH INDEPENDENT-NOISE ESTIMATORS (RECAL.) OR AN AR(1) HISTORY LIKELIHOOD (LIK.; ΔT VERSUS RECALIBRATED SMOOTHING).

κ	Noise	Wait	Fixed	By history	History	Smoothed	Current	Oracle	ΔT_{H-S}
0.4	0.005	80	92.2 / 103.1	90.0 / 100.2	91.6 / 96.2	93.7 / 95.5	93.9 / 103.1	100.0 / 85.1	+0.7 [−0.1, +1.4]
0.4	0.010	80	92.2 / 103.1	90.0 / 100.2	93.2 / 96.8	95.8 / 98.4	95.2 / 113.9	100.0 / 85.1	−1.6 [−2.3, −0.8]
0.5	0.005	100	91.2 / 121.8	90.3 / 119.1	92.6 / 114.3	92.3 / 113.6	90.0 / 119.7	99.2 / 104.7	+0.7 [+0.3, +1.1]
0.5	0.010	100	91.2 / 121.8	90.3 / 119.1	92.2 / 114.9	94.8 / 117.5	95.4 / 131.0	99.2 / 104.7	−2.6 [−3.0, −2.1]
0.6	0.005	120	81.5 / 140.0*	81.5 / 140.0*	80.7 / 134.2*	81.5 / 136.4*	81.5 / 139.6*	81.5 / 125.7	−2.2 [−2.5, −2.0]
0.6	0.010	120	81.5 / 140.0*	81.5 / 140.0*	81.4 / 137.0*	81.5 / 140.0*	81.5 / 140.0*	81.5 / 125.7	−2.9 [−3.3, −2.6]
0.5	AR(1) $\dagger$	100	91.2 / 121.8	90.3 / 119.1	82.0 / 117.3	84.3 / 116.9	93.7 / 123.5	99.2 / 104.7	+0.4
0.5	offset $\dagger$	100	91.2 / 121.8	90.3 / 119.1	97.5 / 117.6	98.8 / 117.9	96.0 / 123.2	99.2 / 104.7	−0.3
0.5	AR(1), recal. $\ddagger$	100	91.2 / 121.8	90.3 / 119.1	88.9 / 117.5*	91.9 / 117.8	93.7 / 123.5	99.2 / 104.7	−0.3
0.5	AR(1) lik. $\ddagger$	—	—	—	94.5 / 117.1	—	—	—	−0.8

On the primary split (500 calibration and 500 evaluation samples), the history-aware rule reached 92.6% success at 114.3 min and the smoothed reporter 92.3% at 113.6 min (Fig. 3(c)). A single fixed wait of 100 min reached 91.2% at 121.8 min, but only 85.0% after the 30-min command (Table III). History-conditioned fixed waits of 92 and 102 min reached 90.3% at 119.1 min, so knowing the previous command explains part of the advantage: the history rule remained 4.8 min faster (paired 95% bootstrap interval 4.0–5.6 min) and the smoothed rule 5.5 min faster (4.7–6.2), against 7.4 min relative to the single wait. Over 20 splits with fresh noise (Fig. 3(d)), the history rule beat the history-conditioned waits in every split (by 4.0–5.6 min), but those waits reached 0.9 evaluation success in only six splits. With every rule calibrated to a shared target of 0.92 (0.94), the conditioned waits reached 0.9 in 18 (20) splits and the history and smoothed rules in all 20, and the history rule stayed faster in every split by a median of 5.6 (6.5) min; on the primary split at 0.92, the conditioned waits (94 and 104 min) and the history rule both reached 92.6%, and the history rule was 6.1 min faster (95% interval 5.3–6.9; success difference 0.0, from −2.1 to 2.2 points; Table III). The gain thus does not rest on a comparator at the edge of the target. The smoothed rule beat the history rule in every split (by 0.2–1.0 min); choosing the fastest feasible threshold gave the same choices, and with fine-grid thresholds frozen on calibration samples smoothing stayed faster (median 0.5–0.8 min at targets 0.90–0.94).

The history estimator depends on its likelihood bandwidth (Fig. 3(b)): with the bandwidth tied to the noise it missed the calibration target at noise s.d. up to 2.5×10^{-3} (69.1–88.6% evaluation success) as its weights collapsed (median effective sample size 1.1 at zero noise versus 28.9 at 0.005), whereas the 0.005 floor reached the target at every level (92.3–92.9%). At doubled noise the current rule became slower than the fixed wait, and Table IV compares history and smoothing at unmatched success (92.2% versus 94.8%). With fine-grid thresholds and smoothing scale frozen on calibration samples at shared targets 0.90–0.94, the history rule was faster in all

20 splits (median 1.4–1.6 min; primary split at 0.92: 1.3 min, 95% interval 0.9–1.8, success difference −0.1 points); the evaluation frontier, used only descriptively, gave 1.3–2.0 min at evaluation success 0.90–0.94 (history thresholds 0.86–0.93, smoothed 0.90–0.94). Under correlated noise that the estimators do not model, the frozen rules lost reliability (smoothed 84.3%, history 82.0%) while staying faster than fixed waiting; a reporter offset made them conservative (97.5–98.8% at about 118 min). Recalibrating the thresholds on AR(1) calibration data restored smoothing (91.9%; 0.9 reached in 19 of 20 splits) but not the history rule (88.9%), whose independent-noise weights stayed overconfident (median effective sample size 8.4 at 90 min), so no threshold met the target. An AR(1) likelihood (covariance $\hat{\sigma}^2 \hat{\rho}^{|a-b|}$, i.e. whitened residuals $(e_i - \hat{\rho}e_{i-1})/\sqrt{1 - \hat{\rho}^2}$ of $e_i = y_i - f_s(t_i)$, $\hat{\rho} = 0.80$ from calibration residuals) raised its success to 94.5% at 117.1 min (5.6 points above recalibration alone, 95% interval 3.8–7.3; effective sample size 163); more reliable than recalibration alone in all 20 splits, it reached 0.9 in only 13 (median 91.5%) but beat the history-conditioned fixed waits (median 89.5%) in every split by 1.4–4.0 min. At $\kappa = 0.6$ even the oracle succeeded for only 81.5% of samples. In this B3 setting feedback saved 4.7–6.5 min (split medians) beyond knowing the previous command at shared calibration targets of 0.90–0.94, smoothing sufficed at the reference noise, and model-based filtering paid off at higher noise and, under correlated noise, only with a likelihood that models the correlation; the results do not establish a waiting benefit from latent memory.

D. Implications for MC Design

In MC terms, a previous command acts on the next one like intersymbol interference whose memory sits in the receiver’s biochemistry rather than only in the channel [6], [8]. A fixed wait is a guard interval: conditioning it on the known previous command lowers the mean wait from 100 to 97 min (92 and 102 min, so the wait after the 30-min command grows), and under pooled calibration the selected waits give unequal per-history success; feedback further adapts to receiver-state

variability within each history, much as adaptive threshold detection adapts to residual interference [9]. As for reception noise in molecular receivers [7], the observation-noise model decides which estimator feedback needs and whether its calibration holds. A prospective test should randomize 3- versus 30-min preconditioning per chamber, probe at 60–120 min, measure the naive response in same-session controls and count sessions rather than cells.

E. Evidence Boundaries

Data: the prediction comparisons are post hoc on inspected roles; in the frozen analysis of the main fresh test, history features alone improved on persistence by only 0.7%; Test C shares a session with inspected data and no experimental-day identifiers exist, so protocol generalization is not cross-day generalization; B3 had seen some protocols. Model: the waiting results come from one simulated model structure whose posterior samples express parameter uncertainty, not validated cell heterogeneity; the 20 splits reuse one finite posterior set rather than independent experiments; the 90% target has no coverage guarantee, and a calibration margin is a sensitivity analysis, not one; the AR(1) parameters came from simulated residuals that an experiment would have to measure; the zero-noise failure of the history estimator reflects finite references, not biology. Task: $\kappa = 0.5$ is a prescribed probe criterion adopted after an 80% version proved infeasible, the naive response is a simulation counterfactual, amplitude and duration were fixed, and the experimental predictors were not tested as decision models, so forecast gains do not imply better waiting decisions.

VI. CONCLUSION

On experimental FGF2–ERK trajectories, reporter filtering and a fixed calibrated mechanistic increment account for most of the 10-min prediction gain, also beyond a lagged non-mechanistic baseline; explicit input history adds little, and the mechanistic increment increased the aggregate error in the evaluated next-command windows. In B3 simulations, feedback-driven waiting saved 4.7–6.5 min beyond fixed waits that use the known previous command at shared calibration targets; smoothing sufficed at the reference noise, while model-based filtering helped at higher noise and, under correlated noise, needed a likelihood that models the correlation.

DATA AND CODE AVAILABILITY

Code, frozen results, protocols with hash records, and verification scripts are available at <https://github.com/ruifeng105/task-aware-cellular-mc>; one command re-runs the analysis and checks the reported values against a frozen reference. Sources are credited in [1], [18], [19]; author files keep their attribution and receive no new license.

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